



Madhav Gupta

Fifth Year Student(2021-2026)

Electrical Eng. Dual Degree(B.Tech + M.Tech)

Indian Institute of Technology Bombay

[linkedin.com/in/madhav-gupta-b68539234/](https://www.linkedin.com/in/madhav-gupta-b68539234/)

Pursuing Minor degree in **Artificial Intelligence and Data Science** offered by CMInDS, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Achieved **99.989 percentile** and All India Rank **164** in **JEE Main** examination amongst 1 million+ students [2021]
- Secured All India Rank **724** out of **0.15 million+** candidates who appeared for **JEE Advanced** examination [2021]
- Recipient of Kishore Vaigyanik Protsahan Yojana (**KVPY**) Fellowship organised by **IISc Bangalore** [2021]
- Recipient of the prestigious National Talent Search Examination (**NTSE**) scholarship, ranked **3rd** in Stage-1 [2019]
- Qualified for Regional Maths Olympiad(**RMO**) by securing a rank in **Top 1%ile** in Pre-RMO by HBCSE [2018]

PUBLICATIONS

- Madhav Gupta**, Vishak Prasad C, and Ganesh Ramakrishnan, "Uncertainty-Aware Subset Selection for Robust Visual Explainability under Distribution Shifts", accepted in **IEEE/CVF WACV 2026** (Algorithms Track)
- Kavitha Viswanathan, **Madhav Gupta**, Amit Sethi et al., "FANVID: A Benchmark for Face and License Plate Recognition in Low-Resolution Videos", *arXiv preprint arXiv:2506.07304*, 2025

PROFESSIONAL EXPERIENCE

Full-Stack AI Intern | NuSummit (National Stock Exchange-IT)

[May '25 - Aug '25]

- Built a POC **RAG-based** Q&A system for financial documents using **Ollama** & **Qdrant**, handling PDFs of 10+ pages
- Integrated end-to-end pipeline using **FastAPI** with user auth and enhanced Angular UI for secure **PDF querying**
- Explored **RAGAS** framework to implement guardrails and enhance **context matching** for accurate response evaluation

Generative AI Solutions Intern | Business Unit-CSW | Hilti Group

[May '24 - Jan '25]

- Integrated 4 apps via APIs, SharePoint, & **Copilot Studio**, boosting info search efficiency by **3x** across platforms
- Implemented **API connectivity** with **topic triggers** to fetch relevant information and generate RAG-style answers
- Presented a successful **POC** to leadership, securing approval for future scaled product development for **10K+** users

AI Research Intern | Sentry AI

[Jan '25 - May '25]

Awarded a **Letter of recommendation** for outstanding work on real-world computer vision problems

- Boosted loitering detection by **50%** via **contrastive learning** for background-invariant clustering across 50+ cameras
- Fine-tuned a Swin Transformer with **K-fold cross-validation** to segregate suitable images, achieving **83%** accuracy
- Slashed false positives by **80%** across 100+ cameras using height-vs-(X,Y) coordinate filtering via **2D linear regression**

Multi-Agent AI Engineer for Business Workflows | Theseus Technologies

[Oct '24 - Nov '24]

- Designed a multi-agent system with **AutoGen** to generate **AI workflows** from **business-specific** inputs and I/O specs
- Orchestrated **subtasks** via a **unified agent** that dynamically spawns or assigns AI agents based on **10+** tool integrations

Conversational AI and LLM Intern | SameSpace

[Nov '23 - Dec '23]

- Built end-to-end **conversational analytics** pipeline spanning data preparation, translation, **fine-tuning**, and inference
- Created **200+** conversations with their **pain points**, agent scores, semantics to fine-tune GPT-2 for their generation
- Translated **OpenOrca** and **Samsam** datasets using **Opus** into Tagalog, fine-tuned a **GPT-2 summarizer** model on them using the **SimpleTransformers** library to generate concise summaries of customer conversations in Tagalog

TECHNICAL & RESEARCH EXPERIENCE

IIT Bombay Racing - Driverless

Part of a 70+ member team designing driverless electric race cars for Formula Student competitions in Germany/UK

Formula Student AI 2023 | Representing the Team at Silverstone

[May '23 - Jul '23]

- Part of the **First ever contingent** of the only Indian team to participate in FS-AI 2023 at the Silverstone Circuit
- Engineered **CAN**(Controller Area Network) interface between **AI** unit and vehicle control unit of **Driverless Car**
- Delivered a **Business Plan Presentation** for an advanced **Level 4 autonomous system** to international judges

Robust Visual Attribution via Submodular Optimization for OOD Data

[Jan '25 - Present]

Thesis Project | Guide: Prof. Ganesh Ramakrishnan

- Built a submodular selection and **perturbation-based uncertainty** framework for enhanced vision model interpretability
- Achieved higher Insertion AUC(**0.91** on synthetic OOD) over prior in-distribution submodular models with **ResNet-101**
- Drafting a research paper for **ICLR CVPR 2026** on robust visual attribution for **Out-of-Distribution** scenarios

Network Inversion for Enhanced NLP Model Interpretability

[Jan '25 - May '25]

R&D Project | Guide: Prof. Amit Sethi

- Designed **Bi-LSTM** and **transformer-based framework** to reconstruct input sequences, boosting model interpretability
- Achieved **93.9%** accuracy in sentiment classification & 63.2% in **sentiment-guided** text generation on Twitter dataset
- Developed custom transformer for **previous token prediction** in LLMs, trained using next token predictor (GPT-2)

Optimizing Object Detection with Detection Transformers (DETR)

[Aug '24 - Nov '24]

Supervised Research Project | Guide: Prof. Amit Sethi

- Evaluated **DETR-based detection**, optimizing backbones (MobileNet-V3, ConvNext, ResNet) on **Pascal VOC** dataset
- Achieved top **mAP (51.4 @50%)** with ConvNext; identified **MobileNet-V3** as optimal for accuracy-efficiency balance
- Fine-tuned encoder-decoder layers and batch sizes to **reduce parameters**, boost mAP, and **mitigate overfitting**

KEY PROJECTS

AI-Powered Tender Discovery and Retrieval System | Industrial Project

[Jan '25 - Mar '25]

Convolution Engineering Consultancy | Transforming tender management using AI

- Built semantic search using **Pinecone**, E5 embeddings and **Reranker** for vector-based ranking across **10K+** tenders
- Implemented **full lifecycle automation**, handling **1K+** new tenders daily, expiry-based deletion and real-time downloads
- Automated **tender scraping** across government portals using **Selenium** generating JSONs with key metadata entities

AI Powered Resume Assistant | Course Project

[Oct '23 - Nov '23]

EE782 - Advanced Machine Learning | Guide: Prof. Amit Sethi

- Fine-tuned a **Mistral-7B** model on the scraped IITB resume repository to capture **domain-specific** writing styles
- Utilized **4-bit quantized** and **sharded models** for parallel inference to optimize GPU usage and fine-tune efficiently

Limestone Data Challenge | Algorithmic Trading Hackathon

[Mar '23 - Apr '23]

Organised by **Finance & Analytics Club**, IIT Bombay in association with **Tower Research Capital**

- Classified 100 stocks into **4** sectors using **hierarchical clustering** and correlation analysis based on their returns data
- Predicted **index returns** using sectoral data, employing **regression** for classification and performance analysis metrics
- Implemented an optimized portfolio allocation strategy with a **Sharpe ratio** over **1.72**, incorporating risk preferences

Statistical and Predictive Analysis | Course Project

[Jan '25 - Apr '25]

DH302 - Introduction to Public Health Informatics | Guide: Prof. Saket Choudhary

- Modeled **2M+** NEET-UG'24 scores with **AIC-scored log-normal** fit; compared with '23 baseline to flag leak anomalies
- Applied **PCA** to 18 amino-acid features from 612 foods & used **hypothesis testing** to validate nutrient-based clusters
- Built **gamma GLM** using AIC on **5+** states to predict monthly conceptions from temperature and humidity trends

TECHNICAL PROFICIENCY AND RELEVANT COURSES

- Programming Languages** : Python, JavaScript, TypeScript, C++, C#, SQL, bash
- Frameworks** : Git, Linux, Angular, .NET Core, Specflow, Selenium, MongoDB, CI/CD pipelines, Copilot Studio
- Libraries** : NumPy, Pandas, Matplotlib, PyTorch, TensorFlow, SciPy, Scikit-Learn, OpenCV, NLTK
- Key Courses** : Programming for Data Science, Data Structure and Algorithms, Advanced Machine Learning, Optimization in Machine Learning, Markov Chains, Probability and Random Processes

POSITION OF RESPONSIBILITY

Interview Coordinator | Placement Office, IIT Bombay

[Aug '22 - Dec '22]

- Coordinated with a team of **250+** members to interview 2000+ students during the **placement season** of 2022-23
- Assisted in conducting Interviews for **20+** firms, handling **student queries**, and maintaining interview records

Event Coordinator | Impulse, EESA's Annual Fest

[Mar '23]

- Acted as a **point of contact** (POC) for top management from companies like **Samsung, Mercedes & Reliance Jio**
- Got a chance to interact with them, fostering **valuable connections** and **enhancing their bonding** with our institution

EXTRA CURRICULAR ACTIVITIES

- Secured **2nd** position in **QuantHive's Algoswarm, Algo-Trading Hackathon**, organised by Finance Club [Aug '23]
- Contributed as a member of the **10-person team** for **DevRev Agent challenge** at **Inter IIT Tech Meet 12.0** [Dec '23]
- Ranked among **top 15** of **250+** teams in Finsearch, presented to **industry professionals** & won cash prize [Aug '22]
- Mentored **10+** participants in **WiDS Bootcamp** on developing a **Sudoku Solver** using **Computer Vision** [Dec '23]
- Achieved **3rd place** in the Electrical Engineering department-wide **table tennis** tournament [Aug '24]